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Enterprise Data Catalog Performance Tuning

Informatica Enterprise Data Catalog Performance Tuning
10.5.6
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Preface

Use the *Enterprise Data Catalog Performance Tuning* guide for information about tuning Enterprise Data Catalog for optimal performance. The performance of Enterprise Data Catalog depends on the size of data that needs to be processed. Tuning the performance involves tuning parameters for metadata ingestion, ingestion database, search, and tuning data profiling.

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To find online support resources on the Informatica Network, visit https://network.informatica.com and select the eSupport option.

CHAPTER 1

Performance Tuning

This chapter includes the following topic:

- Performance Tuning overview, 8

Performance Tuning overview

Performance sizing depends on the volume of data in the catalog, catalog concurrency in terms of concurrent user operations performed in the catalog, and resource concurrency.

You can extract metadata from external sources, such as databases, data warehouses, business glossaries, data integration resources, or business intelligence reports. The performance of Enterprise Data Catalog depends on the size of data being processed. Enterprise Data Catalog classifies data sizes into four categories based on the size of data.

The data sizes are classified into the following categories:

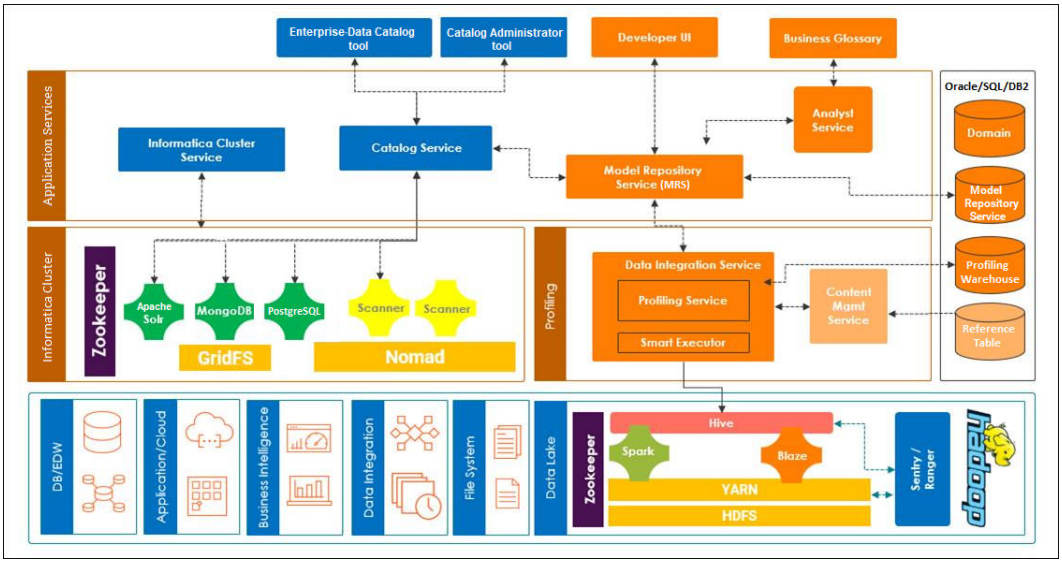
- Low
- Medium
- High
- Very High

Based on the data size, you can configure custom properties in Informatica Administrator to assign predefined parameter values for metadata ingestion, Mongo DB database tuning, and search. Alternatively, you can also individually configure the values for the performance tuning parameters based on your requirement. The profiling tuning involves tuning parameters for the Data Integration Service and the profiling warehouse database properties.

The metadata extraction, storage, search and related operations in an organization include the following phases:

1. Storing the extracted metadata into a catalog for ease of search and retrieval. A catalog represents a centralized repository for storing metadata extracted from different sources. This phase is referred to as the metadata ingestion phase. Enterprise Data Catalog uses Mongo DB as the database for ingesting data.
2. Validating the quality of data with data profiling that includes similarity profiling.
3. Searching for the related data assets in the catalog. Enterprise Data Catalog uses Apache Solr to search for data assets in the catalog.

The following image shows the high-level architecture of Enterprise Data Catalog:



The following table lists the components shown in the architecture diagram:

Stage	Description
Scanner	A scanner is a plug-in component of Enterprise Data Catalog that extracts specific metadata from external data sources.
Data Integration Service	An application service that performs data integration tasks for Enterprise Data Catalog and external applications.
Profiling warehouse	The data store where the profiling information is stored before the profiling information is moved to the catalog.
MongoDB	Enterprise Data Catalog data store for metadata and profiling results.
PostgreSQL	Enterprise Data Catalog data store for value frequency and similarity results.
Apache Solr	Index store for all the extracted metadata for search and retrieval.
Nomad	The service used by Enterprise Data Catalog to manage and schedule scan jobs.
Catalog Service	The application service that runs Enterprise Data Catalog and manages the connections between service components and users.
Informatica Cluster Service	The Informatica Cluster Service runs and manages the services essential to run Enterprise Data Catalog.

CHAPTER 2

Minimum System Requirements

This chapter includes the following topic:

- Minimum system requirements for cluster nodes, 10

Minimum system requirements for cluster nodes

The system requirements listed in the table are for each node in the cluster. Make sure that you allocate the required additional system requirements for the operating system, processes, and other applications running on the node. The values listed are recommendations by Informatica for improving performance. You can increase the values based on your requirements.

System Requirements

System Requirement	Minimum Value
CPU	Eight physical cores with medium clock speed. **Note:** For optimum performance, you must use cores with a minimum clock speed of 2.6 GHz.
Memory	24 GB of unused memory available for use.
Hard disk drive	One high-capacity SSD hard disk drive with 200 GB to 1 TB capacity
File system	ext4 file system.
Network	Bonded Gigabit Ethernet or 10 Gigabit Ethernet between the nodes in the cluster.
Note: For high volume use cases, it is recommended that you have at least 32 GB of unused memory available for the MongoDB database. If the system does not have 32 GB of memory, you must configure 16 GB of swap space.	

You must plan for the following additional system requirements per node based on the features that you plan to use:

MetaDex Server

Additional system requirements to configure the MetaDex server to run as a separate Java Virtual Machine (JVM) on the node that hosts the Informatica domain:

Number of CPU Cores	Memory	Free Disk Space
2	4 GB	20 GB

MetaDex Resource

Additional system requirements to run an MetaDex resource as a separate JVM on the node that hosts the Informatica domain. You need to increase the requirements appropriately based on the volume of metadata that the scanner extracts.

Number of CPU Cores	Memory	Free Disk Space
2	8-32 GB	A minimum of 50 GB

Similarity Discovery

Additional system requirements for the PostgreSQL database used as the data store for value frequency and similarity:

Number of CPU Cores	Memory	Free Disk Space
8	16 GB	A minimum of 200 GB for 2-4 resources that include approximately 10,000 columns in total. You must increase the disk space based on the number of columns.

CHAPTER 3

Sizing Recommendations

This chapter includes the following topic:

- Enterprise Data Catalog sizing recommendations, 12

Enterprise Data Catalog sizing recommendations

Based on the size of data, you must add additional memory and CPU cores to tune the performance of Enterprise Data Catalog. You must also note the minimum number of cores that are required to deploy supported data sizes.

The size of data depends on the number of assets (objects) or the number of datastores. Assets include the sum total of databases, schemas, columns, data domains, reports, mappings and so on in the deployment environment. A datastore represents a repository that stores and manages all the data generated in an enterprise. Datastores include data from applications, files, and databases. Determine if you need to change the default data size for the Enterprise Data Catalog installation. Enterprise Data Catalog has low, medium, high, and very high data sizes or load types Data sizes are classified based on the amount of metadata to process and the number of nodes used to process metadata.

Performance sizing depends on the volume of data in the catalog, catalog concurrency in terms of concurrent user operations performed in the catalog, and resource concurrency.

Note: The minimum recommended load type for production deployment is medium. When the number of supported concurrent users for a load type increases, it is recommended that you move to the next higher load type.

Based on the parameters that affect performance sizing, you can plan to move to the next load type as required.

If you increase the number of nodes, you must perform the following steps:

1. Add the **Shared File Path** property in Informatica Cluster Service configuration.
2. Restart the Informatica Cluster Service.
3. Restart the Catalog Service.
4. Re-index the catalog.

Low

Low represents two million assets or 30-40 datastores. The maximum user concurrency to access Enterprise Data Catalog is 20 users and the recommended resource concurrency is 1 medium data size resource or 2

low data size resources. It is not advisable to run high data size resources that require high memory allocation for low load type use cases.

For a low load type, it is recommended that you plan for one node to run the Informatica domain, one node for Enterprise Data Catalog, and an additional node to run similarity.

System Requirements

The following table lists the recommended minimum system requirements for application services and infrastructure for low load type:

Component	CPU Cores	Memory	Storage
Infrastructure*	8	16 GB	200 GB
Data Integration Service+	8	16 GB	20 GB
Profiling warehouse database server	4	16 GB	50 GB
MetaDex server	2	4 GB	50 GB
One MetaDex resource	2	8-32 GB	50 GB
Single node cluster for services associated with Informatica Cluster Service	8	24 GB	120 GB
Similarity profiling node	8	16 GB	200 GB

*Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, Similarity profiling data store database, and Informatica Cluster Service.

+The hardware sizing requirement listed is for a use case that includes profiling.
Note: You can run the Data Integration Service on the same node as the domain or on a separate node. If the Data Integration Service used for profiling co-exists on the Infrastructure node, the total hardware sizing is the sum of Data Integration Service and Infrastructure hardware components.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the CPU requirements listed in the table by 1.5 to 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 5 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

Note: The Catalog Service heap size is 2 GB. The maximum heap size for Data Integration Service is 2 GB.

Service Level Agreement

The Service Level Agreement (SLA) to process metadata extraction and profiling for a low data size is approximately two business days. The SLA calculation is for a basic RDBMS resource such as Oracle.

Calculation

The complexity of transformations present in the extracted metadata affect the memory required for lineage. Ensure that you increase the memory for infrastructure as required.

Medium

Medium represents 20 million assets or 200-400 datastores. The maximum user concurrency to access Enterprise Data Catalog is 50 and the recommended resource concurrency is 3 medium data size resources or 1 high data size resource that requires high memory allocation. A medium data size resource requires a minimum of three nodes to run the cluster. You configure all the three nodes as data nodes and processing nodes.

For a medium load type, it is recommended that you plan for one node to run the Informatica domain, three nodes to run Enterprise Data Catalog and an additional node to run similarity.

You can choose to use the minimum system requirements or system requirements for a high volume source as necessary. The system requirement tables list the components and values for infrastructure, Data Integration Service, and profiling warehouse.

Minimum System Requirements

The following table lists the recommended minimum system requirements for application services and infrastructure for medium load type:

Component	CPU Cores	Memory	Storage
Infrastructure*	24	32 GB	200 GB
Data Integration Service+	32	64 GB	100 GB
Profiling warehouse database server	8	32 GB	200 GB
MetaDex server	4	4 GB	50 GB
One MetaDex resource	2	8-32 GB	100 GB
Requirements per node for Informatica Cluster Service and associated services	8	24 GB	600 GB
Total requirements for three nodes for Informatica Cluster Service and associated services	24	72 GB	1.8 TB

Component	CPU Cores	Memory	Storage
Similarity discovery profiling node	8	16 GB	500 GB

* Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, and Informatica Cluster Service.

+The hardware sizing requirement listed is for a use case that includes profiling.
Note: You can run the Data Integration Service on the same node as the domain or on a separate node. If the Data Integration Service used for profiling co-exists on the Infrastructure node, the total hardware sizing is the sum of Data Integration Service and Infrastructure hardware components.

For better stability and throughput, it is recommended that you configure an additional processing node for the Informatica Cluster Service. To support three high memory resources where 12 GB scanner memory is configured, you must allocate at least 48 GB of system memory for the additional node.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the CPU requirements listed in the table by 1.5 to 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 20 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

Note: The Catalog Service heap size is 4 GB. The maximum heap size for Data Integration Service is 10 GB.

Service Level Agreement

The following table describes the Service Level Agreement (SLA) to process metadata extraction and profiling for medium load type:

Process	SLA
Metadata extraction	Approximately one business week to accumulate 20 million assets based on metadata source types. The SLA calculation is for a basic RDBMS resource such as Oracle or PowerCenter.
Profiling	Approximately one business week in addition to metadata extraction SLA. The SLA calculation is for a basic RDBMS resource, such as Oracle, with 10K sampling rows for profiling and 10 data domains. **Note:** The time taken to complete profiling is linearly proportional to the total number of data domains selected for profiling.

Calculation

When you calculate the system requirements, make sure that you add the system requirements for infrastructure and three cluster nodes, and as required for profiling and Catalog Service tuning.

Recommended System Requirements for High Volume Sources

A high volume source might include a single large resource or large number of resources. As a general guideline, if the object count in a resource exceeds the recommended limit for medium scanner type, the resource is considered to be a high volume source. For more information about the object count, see "Metadata extraction scanner memory parameters" on page 23.

System Requirements

The following table lists the system requirements for application services and infrastructure for high volume sources for the medium load type:

Component	CPU Cores	Memory	Storage
Infrastructure	24	32 GB	200 GB
Data Integration Service[+]	64	128 GB	100 GB
Profiling warehouse database server	8	32 GB	200 GB
MetaDex server	4	4 GB	50 GB
One MetaDex resource	2	8-32 GB	100 GB
Requirements per node for Informatica Cluster Service and associated services	12	48 GB	600 GB
Total requirements for three nodes for Informatica Cluster Service and associated services	36	144 GB	1.8 TB
Similarity discovery profiling node	8	16 GB	1 TB

* Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, and Informatica Cluster Service. You can configure one Data Integration Service or configure Data Integration Service on two nodes where each node includes 32 CPU cores and 64 GB of memory.

[+]The hardware sizing requirement listed is for a use case that includes profiling.
Note: You can run the Data Integration Service on the same node as the domain or on a separate node. If the Data Integration Service used for profiling co-exists on the Infrastructure node, the total hardware sizing is the sum of Data Integration Service and Infrastructure hardware components.

For better stability and throughput, it is recommended that you configure an additional processing node for the Informatica Cluster Service. To support three high memory resources where 12 GB scanner memory is configured, you must allocate at least 48 GB of system memory for the additional node.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the CPU requirements listed in the table by 1.5 to 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 20 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

For more information about high volume sources, see the "High volume sources" on page 22 topic.

Service Level Agreement

The following table describes the SLA to process metadata extraction and profiling for medium load type for high volume sources:

Process	SLA
Metadata extraction	Approximately three business days to accumulate 20 million assets based on metadata source types. The SLA calculation is for a basic RDBMS resource such as Oracle or PowerCenter.
Profiling	Approximately four business days in addition to metadata extraction SLA. The SLA calculation is for a basic RDBMS resource, such as Oracle, with 10K sampling rows for profiling and 10 data domains. **Note:** The time taken to complete profiling is linearly proportional to the total number of data domains selected for profiling.

The throughput or performance depends on the number of columns in the table, file size, file type, complexity of mapping, size of the reports, file content, and number of data domains for profiling.

High

High represents 50 million assets or 500-1000 data stores. The maximum user concurrency to access Enterprise Data Catalog is 75 and the recommended resource concurrency is 5 medium data size resources or 1 high data size resource that requires high memory allocation. A high data size resource is recommended to be run on 6 nodes.

You can choose to use the minimum system requirements or system requirements for a high volume source as necessary. The system requirement tables list the components and values for infrastructure, Data Integration Service, and profiling warehouse.

Minimum System Requirements for Existing High Load Type Configuration Users

As an existing user, you must plan for an additional node to run similarity. If the Informatica Cluster Service runs on a cluster with six nodes and you plan to upgrade, see the following table for the minimum system requirements:

Component	CPU Cores	Memory	Storage
Infrastructure*	48	64 GB	300 GB
Data Integration Service+	32	64 GB	500 GB
Profiling warehouse database server	16	64 GB	500 GB
MetaDex server	8	4 GB	50 GB
One MetaDex resource	2	8-32 GB	100 GB
Requirements per node for Informatica Cluster Service and associated services	8	24 GB	1 TB

Component	CPU Cores	Memory	Storage
Total requirements for six nodes for Informatica Cluster Service and associated services	48	144 GB	6 TB
Similarity discovery profiling node	8	16 GB	1 TB

* Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, and Informatica Cluster Service. You can configure one Data Integration Service or configure Data Integration Service on two nodes where each node includes 32 CPU cores and 64 GB of memory.

+The hardware sizing requirement listed is for a use case that includes profiling.
Note: You can run the Data Integration Service on the same node as the domain or on a separate node. If the Data Integration Service used for profiling co-exists on the Infrastructure node, the total hardware sizing is the sum of Data Integration Service and Infrastructure hardware components.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the CPU requirements listed in the table by 1.5 to 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 100 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

Note: The Catalog Service heap size is 9 GB. The maximum heap size for Data Integration Service is 20 GB.

Service Level Agreement

The following table describes the SLA to process metadata extraction and profiling for high load type:

Process	SLA
Metadata extraction	Approximately two business weeks to accumulate 50 million assets based on metadata source types. The SLA calculation is for a basic RDBMS resource such as Oracle or PowerCenter.
Profiling	Approximately two business weeks in addition to metadata extraction SLA. The SLA calculation is for a basic RDBMS resource, such as Oracle, with 10K sampling rows for profiling and 10 data domains. **Note:** The time taken to complete profiling is linearly proportional to the total number of data domains selected for profiling.

Calculation

When you calculate the system requirements, make sure that you add the system requirements for infrastructure and six cluster nodes, and as required for profiling and Catalog Service tuning. For better throughput with additional hardware capacity, see the performance tuning section for each component in this article.

Recommended System Requirements for New Users and Existing Medium Load Type Configuration Users Planning to Move to a Higher Load Type

If you are a new user or an existing user planning to move from an existing load type to a higher load type, see the listed system requirements.

System Requirements

The table lists the recommended system requirements for application services and infrastructure for the following scenarios:

- You are a new user with Informatica Cluster Service running on a three-node cluster with the load type configured as high.
- You are an existing user with Informatica Cluster Service running on a three-node cluster with the load type configured as medium. You plan to upgrade to version 10.5.1 and configure a higher load type.

Component	CPU Cores	Memory	Storage
Infrastructure	48	64 GB	300 GB
Data Integration Service[+]	32	64 GB	500 GB
Profiling warehouse database server	16	64 GB	500 GB
MetaDex server	8	4 GB	50 GB
One MetaDex resource	2	8-32 GB	100 GB
Requirements per node for Informatica Cluster Service and associated services	16	48 GB	2 TB
Total requirements for three nodes for Informatica Cluster Service and associated services	48	144 GB	6 TB
Similarity discovery profiling node	8	16 GB	1 TB

* Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, and Informatica Cluster Service. You can configure one Data Integration Service or configure Data Integration Service on two nodes where each node includes 32 CPU cores and 64 GB of memory.

[1] Indicates that you can use the specified memory for Data Integration Service, or use two Data Integration Service nodes of 32 cores and 64 GB each. The hardware sizing requirement listed is for a use case that includes profiling.

If you run the Informatica Cluster Service on a three-node cluster with the load type configured as high, for better stability and throughput, it is recommended that you configure an additional processing node for the Informatica Cluster Service.

To support three high memory resources where 12 GB scanner memory is configured, you must allocate at least 48 GB of system memory for the additional node.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the CPU requirements listed in the table by 1.5 to 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 100 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

You can configure all the three service nodes as data nodes and processing nodes.

For more information about high volume sources, see "High volume sources" on page 22.

Service Level Agreement

The following table describes the SLA to process metadata extraction and profiling for high volume sources for the high load type:

Process	SLA
Metadata extraction	Approximately one-two business weeks to accumulate 50 million assets based on metadata source types. The SLA calculation is for a basic RDBMS resource such as Oracle or PowerCenter.
Profiling	Approximately two business weeks in addition to metadata extraction SLA. The SLA calculation is for a basic RDBMS resource, such as Oracle, with 10K sampling rows for profiling and 10 data domains. **Note:** The time taken to complete profiling is linearly proportional to the total number of data domains selected for profiling.

The throughput or performance depends on the number of columns in the table, file size, file type, complexity of mapping, size of the reports, file content, and number of data domains for profiling.

Recommendations

You can consider the following recommendation when you want to optimize performance for high volume sources in the high load type object limit:

- To increase the profiling throughput and use the recommended hardware effectively, see "setConcurrencyParameters Script" on page 45 to tune the parameters as necessary.

Very High

Very high represents 100 million assets or 500-1000 data stores with high volume sources. The maximum user concurrency to access Enterprise Data Catalog is 100 and the recommended resource concurrency is 5 medium data size resources or 1 high data size resource that requires high memory allocation. A very high data size resource is recommended to be run on 6 nodes.

For a very high load type, it is recommended to plan for one node for the Informatica domain, six nodes to run Enterprise Data Catalog, and an additional node to run similarity. Out of the six nodes on which you run Enterprise Data Catalog, you can configure three nodes as data nodes and three nodes as processing nodes.

The system requirement tables list the components and values for infrastructure, Data Integration Service, and profiling warehouse.

Minimum System Requirements

The following table lists the recommended minimum system requirements for application services and infrastructure:

Component	CPU Cores	Memory	Storage
Infrastructure*	48	64 GB	300 GB
Data Integration Service+	64	128 GB	500 GB

Component	CPU Cores	Memory	Storage
Profiling warehouse database server	32	64 GB	500 GB
MetaDex server	8	4 GB	50 GB
One MetaDex resource	2	8-32 GB	100 GB
Requirements per node for Informatica Cluster Service and associated services	16	48 GB	2 TB
Total requirements for six nodes for Informatica Cluster Service and associated services	96	288 GB	12 TB
Similarity discovery profiling node	8	16 GB	2 TB

* Includes Informatica domain services, Model Repository Service, Catalog Service, and Content Management Service, and Informatica Cluster Service. You can configure one Data Integration Service or configure Data Integration Service on two nodes where each node includes 32 CPU cores and 64 GB of memory.

†The hardware sizing requirement listed is for a use case that includes profiling.
Note: You can run the Data Integration Service on the same node as the domain or on a separate node. If the Data Integration Service used for profiling co-exists on the Infrastructure node, the total hardware sizing is the sum of Data Integration Service and Infrastructure hardware components.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the CPU requirements listed in the table by 1.5 to 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 200 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

Note: The Catalog Service heap size is 18 GB. The maximum heap size for Data Integration Service is 20 GB.

Service Level Agreement

The following table describes the SLA to process metadata extraction and profiling for high load type:

Process	SLA
Metadata extraction	Approximately two-four business weeks to accumulate 100 million assets based on metadata source types. The SLA calculation is for a basic RDBMS resource such as Oracle or PowerCenter.
Profiling	Approximately two business weeks in addition to metadata extraction SLA. The SLA calculation is for a basic RDBMS resource, such as Oracle, with 10K sampling rows for profiling and 10 data domains. **Note:** The time taken to complete profiling is linearly proportional to the total number of data domains selected for profiling.

Calculation

When you calculate the system requirements, make sure that you add the system requirements for infrastructure and six cluster nodes, and as required for profiling and Catalog Service tuning. For better

throughput with additional hardware capacity, see the performance tuning section for each component in this article.

High volume sources

A high volume source might include a single large resource or large number of resources. As a general guideline, if the object count in a resource exceeds the recommended limit for medium scanner type, the resource is considered to be a high volume source. For more information about the object count, see "Metadata extraction scanner memory parameters" on page 23.

Recommendations to Optimize the Performance

You can tune the components to optimize the performance when you scan a high volume source, or when one or more of the following use cases is true:

- Re-indexing a large catalog.
- Re-assigning the connection assignment parameters.
- Deleting a large resource.
- Scanning a large PowerCenter repository.
- Scanning a large-volume business intelligence report.
- Scanning an Oracle source with large volume public synonym.
- Increasing upgrade throughput when you move to a higher version.
- Profiling millions of unstructured files.

To optimize the performance, you can consider the following recommendations:

- Split a large resource into multiple resources, if possible.
- Increase the disk space by 10 GB for every additional 1 million assets.
- When a single large resource exceeds the **High** scanner memory option limit, you can increase the memory for the Catalog Service using the custom properties for low, medium, or high based on your requirement. For example, in a high load type deployment, you can increase the memory using the **LdmCustomOptions.scanner.memory.high** value as necessary in the **Informatica Administrator** > **Manage** > **Services and Nodes** > **Catalog Service** > **Custom Properties** section.

  For example, when you want to run complex mappings based on large resources, large reports, or large unstructured files, or if you want to include all the schemas in a resource scan, then increase the custom property value.

  Alternatively, you can use the *-Dscanner* JVM option to increase the memory for the specific resource when you configure the resource.
- When you want to run a profile on millions of unstructured files, increase the node process heap size to 2 GB.

  To increase the node process heap size, perform the following steps:

  1. Open the **infaservice.sh** file located in the `<Informatica installation directory>/tomcat/bin/` directory.
  2. Increase the value of the *INFA_JAVA_OPTS* parameter from **INFA_JAVA_OPTS="-Xmx1024m"** to **INFA_JAVA_OPTS="-Xmx2048m"**.
  3. Save the file.
  4. Restart the Informatica domain.

- When the number of files and tables in a resource exceed 100,000, increase the Model Repository Service heap size to 4 GB.

  To increase the heap size, navigate to **Informatica Adminstrator** > **Services and Nodes** > **Model Repository Service** > **Properties** > **Advanced Properties** section and set **Maximum Heap Size** to **2048m**.

Metadata extraction scanner memory parameters

Depending on the size of source data for a resource, you can use one of the following parameters to configure the memory requirements for the scanner to extract metadata:

Parameter	Memory Option*	Supported Object Count for Default Memory[1]	Default Memory[2]
LdmCustomOptions.scanner.memory.low	low	Up to 1000 tables Up to 2000 files Up to 100 mappings/reports Up to 100,000 objects Up to a maximum file size of 5 MB for unstructured files	2 GB
LdmCustomOptions.scanner.memory.medium	medium	Up to 10,000 tables Up to 20,000 files Up to 1000 mappings/reports Up to 1M objects Up to a maximum file size of 10 MB for unstructured files	4 GB
LdmCustomOptions.scanner.memory.high	high	Up to 100,000 tables Up to 1,000,000 files Up to 20,000 Mappings/Reports Up to 10M objects Up to a maximum file size of 100 MB for unstructured files	12 GB

* Indicates the memory allocated to scanner process heap size.

[1] Depends on the number of columns in the table, file size, file type, complexity of the mapping, size of the report, and object content.

[2] Indicates the default values configured for the scanner based on the data size.

Increase the value of the **LdmCustomOptions.scanner.memory.high** custom property as necessary or configure the *-Dscanner.memory.high=<Memory in MB> JVM* option when one of the following conditions is true:

- You want to run complex mappings based on large resources, large reports, or large unstructured files.
- You want to include all the schemas in a resource scan.
- A single large resource exceeds the High scanner memory option limit.

To increase the value of the **LdmCustomOptions.scanner.memory.high** custom property, perform the following steps:

1. In Informatica Administrator, navigate to the **Manage** > **Services and Nodes** > **Catalog Service** > **Custom Properties** section.

2. Edit the property, enter the required value, and save the property.

To configure the *-Dscanner.memory.high=<Memory in MB> JVM* option, perform the following steps:

1. In Catalog Administrator, edit the resource.
2. In the **Metadata Load Settings** > **Advanced Properties** section, configure the following options:
   - Memory. Choose **High**.
   - JVM Options. Enter **-Dscanner.memory.high=<Memory in MB>**.
3. Save and run the resource.

Note:

- You must select the number of concurrent scanners based on the memory type and the available resources in the cluster.
- You must increase the Informatica PowerCenter scanner memory based on the complexity of the mappings.

Database Management Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from database resources based on the resource memory configured:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Oracle	Table	1K	100K
MS SQL	Table	1K	100K
SAP HANA	Table	5K	50K
Teradata	Table or View	1K	10K
MS SQL Stored Procedure	Stored Procedure	5K	5K

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Oracle	Table	10K	1 million
MS SQL	Table	10K	1 million
SAP HANA	Table	10K	100K
Teradata	Table or View	5K	1 million
MS SQL Stored Procedure	Stored Procedure	10K	10K

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Oracle	Table	100K	5 million
MS SQL	Table	100K	5 million
SAP HANA	Table	20K	200K
Teradata	Table or View	20K	5 million
MS SQL Stored Procedure	Stored Procedure	20K	20K

MetaDex Database Stored Procedure Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from MetaDex database stored procedure resources based on the resource memory configured:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
Oracle	Stored procedure	500	15K
Oracle With automatic entry point detection and recursive calls enabled	Stored procedure	1K root assets 1K child assets	30K
Oracle Script	Script	500	10K
MS SQL	Stored procedure or view	500	25K
Teradata	Stored procedure	500	15K
Teradata Bteq Script	Script	500	10K

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
Oracle	Stored procedure	1K	50K
Oracle With automatic entry point detection and recursive calls enabled	Stored procedure	2K root assets 2K child assets	60K

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
Oracle Script	Script	1K	20K
MS SQL	Stored procedure or view	1K	50K
Teradata	Stored procedure	1K	50K
Teradata Bteq Script	Script	2K	50K

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
Oracle	Stored procedure	5K	125K
Oracle With automatic entry point detection and recursive calls enabled	Stored procedure	5K root assets 5K child assets	120K
Oracle Script	Script	5K	100K
MS SQL	Stored procedure or view	2K	100K
Teradata	Stored procedure	5K	125K
Teradata Bteq Script	Script	10K	150K

Data Integration Resources Memory Recommendations

The following load type tables list the maximum number of assets that you can extract from data integration resources based on the resource memory configured:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
PowerCenter	Mapping	1K	100K
IBM DataStage	Mapping	1K	10K
Oracle Data Integrator	Mapping	10	1K
IICS	Mapping	5K	100K
Big Data Management	Mapping	1K	1 million

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Big Data Streaming	Mapping	500	100K
SSIS	Mapping or File	100	100K
Custom Lineage	Links	100K	-
Database Scripts	Scripts	500	500

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
PowerCenter	Mapping	10K	1 million
IBM DataStage	Mapping	5K	50K
Oracle Data Integrator	Mapping	100	10K
IICS	Mapping	10K	1 million
Big Data Management	Mapping	5K	5 million
Big Data Streaming	Mapping	1K	250K
SSIS	Mapping or File	500	2 million
Custom Lineage	Links	250K	-
Database Scripts	Scripts	1K	1K

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
PowerCenter	Mapping	25K	5 million
IBM DataStage	Mapping	10K	100K
Oracle Data Integrator	Mapping	250	50K
IICS	Mapping	20K	2 million
Big Data Management	Mapping	10K	10 million
Big Data Streaming	Mapping	2K	500K
SSIS	Mapping or File	1K	5 million

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Custom Lineage	Links	1 million	-
Database Scripts	Scripts	5K	5K

MetaDex Data Integration Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from the MetaDex data integration resources based on the resource memory configured:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
SSIS	Data Flow	250	12.5K
Talend Data Integration	Mapping	500	10K
Oracle Data Integrator	Scenarios	1K	10K
IBM DataStage	Jobs	250	25K
SAP BODS	Data Flow	500	25K
IICS V2	Mapping	50	40K

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
SSIS	Data Flow	500	25K
Talend Data Integration	Mapping	1K	20K
Oracle Data Integrator	Scenarios	2K	20K
IBM DataStage	Jobs	500	50K
SAP BODS	Data Flow	1K	50K
IICS V2	Mapping	100	80K

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
SSIS	Data Flow	1K	50K
Talend Data Integration	Mapping	2K	5K
Oracle Data Integrator	Scenarios	5K	50K
IBM DataStage	Jobs	2.5K	250K
SAP BODS	Data Flow	4K	200K
IICS V2	Mapping	500	160K

Business Intelligence Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from business intelligence resources based on the resource memory configured:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
MicroStrategy	Reports	500	10K
Qlik Sense	Reports	500K	500K
SAP BusinessObjects	Reports	100	25K `
IBM Cognos	Reports	250	100K

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
MicroStrategy	Reports	1K	50K
Qlik Sense	Reports	1 million	1 million
SAP BusinessObjects	Reports	1K	500K
IBM Cognos	Reports	500	180K

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
MicroStrategy	Reports	5K	100K
Qlik Sense	Reports	5 million	5 million
SAP BusinessObjects	Reports	5K	1 million
IBM Cognos	Reports	1K	240K

Custom Scanner Resource Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from a custom scanner resource based on the resource memory configured:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Custom scanner	Objects	5K	5K

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Custom scanner	Objects	10K	10K

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Custom scanner	Objects	20K	20K

No SQL Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from No SQL resources based on the resource memory configured:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
MongoDB	Collections	1K	100K
Apache Cassandra	Table	1K	100K

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
MongoDB	Collections	5K	1 million
Apache Cassandra	Table	10K	1 million

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
MongoDB	Collections	10K	5 million
Apache Cassandra	Table	100K	5 million

Cloud Data Warehouse Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from cloud data warehouse resources based on the resource memory configured:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Google BigQuery	Table	1K	1 million
Snowflake	Table	1K	100K
Amazon Redshift	Table	1K	100K
ADLS	File	10K	10K
Amazon S3	File	10K	10K

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Google Cloud Storage	File	10K	10K
AWS Glue	Object	100K	1 million

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Google BigQuery	Table	5K	5 million
Snowflake	Table	5K	500K
Amazon Redshift	Table	5K	500K
ADLS	File	50K	10K
Amazon S3	File	50K	10K
Google Cloud Storage	File	50K	10K
AWS Glue	Object	250K	5 million

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Google BigQuery	Table	10K	10 million
Snowflake	Table	10K	1 million
Amazon Redshift	Table	10K	1 million
ADLS	File	100K	10K
Amazon S3	File	100K	10K
Google Cloud Storage	File	100K	10K
AWS Glue	Object	500K	2 million

The following load type tables list the maximum number of assets that you can extract in schema merge from cloud data warehouse resources based on the resource memory configured:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	Folder Size	Enterprise Data Catalog Asset Count
Amazon S3	File	10 GB	15K

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	Advanced Scanners Asset Count	Enterprise Data Catalog Asset Count
Amazon S3	File	20 GB	30K

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	Advanced Scanners Asset Count	Enterprise Data Catalog Asset Count
Amazon S3	File	50 GB	60K

Streaming Resource Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from streaming resource based on the resource memory configured:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Apache Kafka	Topics	1K	25K

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Apache Kafka	Topics	5K	50K

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Apache Kafka	Topics	10K	100K

Data Engineering Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from data engineering resources based on the resource memory configured:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Apache Atlas	Links	5K	5K
Cloudera Navigator	Entities	5K	5K
Apache Hive	Tables	1K	100K
HDFS	Files	1K	10K

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Apache Atlas	Links	25K	25K
Cloudera Navigator	Entities	10K	10K
Apache Hive	Tables	10K	1 million
HDFS	Files	10K	1 million

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	Data Type Count	Maximum Asset Count
Apache Atlas	Links	50K	50K
Cloudera Navigator	Entities	20K	20K
Apache Hive	Tables	50K	5 million
HDFS	Files	100K	5 million

MetaDex Data Analytics and Reporting Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from MetaDex data analytics and reporting resources based on the resource memory configured:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
SAS	Scripts	250	250K
SSAS	Analysis files	1K	200K
SSRS	Reports	1K	15K
Snowflake	Tables	1K	200K
Microsoft Power BI	Reports	50	30K
Microsoft Azure Data Factory	Pipelines	100	25K

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
SAS	Scripts	500	500K
SSAS	Analysis files	2K	400K
SSRS	Reports	2K	30K
Snowflake	Tables	2K	400K
Microsoft Power BI	Reports	100	60K
Microsoft Azure Data Factory	Pipelines	500	100K

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
SAS	Scripts	1K	1 million
SSAS	Analysis files	5K	1 million
SSRS	Reports	5K	100K
Snowflake	Tables	10K	2 million
Microsoft Power BI	Reports	250	150K
Microsoft Azure Data Factory	Pipelines	1K	200K

MetaDex ETL Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from MetaDex ETL resources based on the resource memory configured:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
Azure Databricks Notebooks	Notebook	100	7K

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
Azure Databricks Notebooks	Notebook	500	35K

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
Azure Databricks Notebooks	Notebook	1K	70K

MetaDex Legacy Resources Memory Recommendation

The following load type tables list the maximum number of assets that you can extract from MetaDex legacy resources based on the resource memory configured:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
COBOL scripts	Scripts	250	2.5K
COBOL programs using JCL scripts	Scripts	250	10K

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
COBOL scripts	Scripts	500	5K
COBOL programs using JCL scripts	Scripts	500	25K

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	MetaDex Asset Count	Enterprise Data Catalog Asset Count
COBOL scripts	Scripts	1K	10K
COBOL programs using JCL scripts	Scripts	1K	100K

Tuning configuration parameters for resources

You can configure parameters to optimize the performance of resources.

Cloud Resources

You can tune the performance parameters for cloud resources such as Azure Data Lake Storage and Google Cloud Storage resources.

To increase the source metadata extraction throughput, tune the **-DfileProcessorCount** JVM option. By default, this option is set to **2** for all file-based resources.

You can configure the **-DfileProcessorCount** parameter based on the number of cores required for a node in the cluster as shown in the following table:

Number Of Cores Required for a Node In The Cluster	JVM Option and Value
8	DfileProcessorCount=8
16	DfileProcessorCount=16

To add, edit, or delete a JVM option, navigate to the **Metadata Load Settings** tab of the resource in Catalog Administrator.

Tuning performance for similarity

Enterprise Data Catalog stores the similarity and value frequency details in the PostgreSQL database.

Similarity discovery between resources is calculated based on column name, data pattern in the column, and the unique values in the columns. As similarity is a resource-intensive process, it is recommended that you plan for a disk space based on the data volume. The minimum recommendation for disk space is 200 GB to support a similarity resource group with two resources and a total of approximately 10,000 columns in total.

The following table lists the heap size recommendations for similarity resource groups:

Load Type	Heap Memory	Total Number of Colums per Similarity Resource Group
Low	2 GB	Up to 10K columns.
Medium	4 GB	Up to 100K columns.

You can tune the computation throughput by adding additional hardware for the PostgreSQL database. You can configure the additional hardware using the `LdmCustomOptions.scanner.simDiscoveryScanner.partitionPairJobCount` Catalog Service custom

property. Default value for the property is 5 for a PostgreSQL node that includes 8 CPU cores and 16 GB of memory. To increase the value by 5 for the custom property, you must add 4 CPU cores and 4 GB of memory.

The recommended number of resources in a similarity resource group is 4. Based on the number of columns in a resource group and the available hardware, you can use the `LdmCustomOptions.scanner.simDiscoveryScanner.maxPermittedResourceCount` Catalog Service custom property to specify the number of resources in a similarity resource group.

Enterprise Data Catalog agent

You can use Enterprise Data Catalog Agent to extract metadata from data sources that run on Microsoft Windows.

The following table lists the recommended system requirements to deploy Enterprise Data Catalog Agent:

Component	Minimum System Requirements
CPU Cores	8
Memory	16 GB

For more information, see Appendix D: Enterprise Data Catalog Agent in the *Catalog Administrator Guide*.

Log files

To monitor the performance and to identify issues, you can view the log files. Log files are generated at every stage of metadata extraction and ingestion.

You can view the following log files for Enterprise Data Catalog:

Accessing the Log files from the UI

You can access the log files from the following Enterprise Data Catalog URL: `http://<host name>:<Catalog Service port>/access/ui/index`

Catalog Service Log File

Navigate to the `<Informatica installation directory>/logs/<node_name>/services/CatalogService/<Catalog_service_name>/LDM.log` directory to view the Catalog Service log file.

Scanner Log File

In Catalog Administrator, you can configure the scanner log level for a resource and MITI log level for data sources that support MITI. You can set the log level as NONE, FATAL, ERROR, WARNING, STATUS, INFO, or DEBUG.

To configure the log level, perform the following steps:

- To configure the scanner log level, navigate to the resource and set the following JVM option in the **Metadata Load Settings** > **Advanced Properties** section:
  `-Dscannerloglevel=<scanner_log_level>`

  For example, `-Dscannerloglevel=INFO`
- To configure the MITI log level, navigate to the resource and set the following JVM option in the **Metadata Load Settings** > **Advanced Properties** section:
  `-DagentLogLevel=<scanner_log_level>`

  For example, `-DagentLogLevel=INFO`.

Note: By default, the scanner log level is INFO and log level for data sources supported by MITI is DEBUG. It is recommended to reduce the log level when the log file size increases in GB during a single large resource run.

In Catalog Administrator, you can view the Log URL for resources in the **Monitoring** > **Tasks** section.

MongoDB Log Files

You can access the MongoDB log files from the directory that you configured for the **MongoDB Log Directory** property when you created the Informatica Cluster Service. Default is `<Cluster custom directory>/mongo/log`.

Apache ZooKeeper Log Files

You can access the Apache ZooKeeper log files from the directory that you configured for the **ZooKeeper Data Directory** property when you created the Informatica Cluster Service. Default is `$clusterCustomDir/zk/data`.

Apache Solr Log Files

You can access the Apache Solr log files from the directory that you configured for the **Solr Data Directory** property when you created the Informatica Cluster Service. Default is `<Cluster custom directory> /solr/data`.

Nomad Aggregation Log Files

You can view the Nomad aggregation log files for high volume load types using the `nomad alloc logs -f <Task_ID> > /path` command. You can get the <Task_ID> from the Nomad UI.

PostgreSQL Log Files

You can access the PostgreSQL log files from the directory that you configured for the **PostgreSQL DB Log Directory** property when you created the Informatica Cluster Service. Default is `<Cluster custom directory>/postgres/log`.

Data Integration Service Log File

To view the log file, navigate to the `<Informatica installation directory>/logs/<node_name>/services/DataIntegrationService/disLogs/dis_<dis_ service_name>_<node_name>.log.0` directory.

Mapping Log File

To view the log file, navigate to the `<Informatica installation directory>/logs/<node_name>/services/DataIntegrationService/disLogs/profiling/<logs>` directory.

XMI File Logs

You can use the following flag when you configure a resource to retain the XMI log files: `-DpersistXmiLocation=<path in the cluster>`

CHAPTER 4

Performance Tuning Recommendations

This chapter includes the following topic:

- Performance tuning for profiling, 40

Performance tuning for profiling

Tuning profiling performance involves configuring the Data Integration Service parameters, Data Integration Service concurrency parameters, and profiling warehouse parameters.

Tuning Data Integration Service parameters for profiling

To optimize data profiling, you can configure Data Integration Service parameters, Data Integration Service profiling tuning parameters, and Data Integration Service grid configuration. You can also tune unique key inference performance parameters.

Note: Performance and stability issues are noted when you configure Data Integration Service and run profiles on a grid. For optimal performance, configure Data Integration Service on a standalone node or multiple nodes with horizontal scaling instead of a grid.

You need to configure the Temporary Directories and Maximum Execution Pool Size parameters for the Data Integration Service. You can configure parameters, such as Reserved Profile Threads that apply to the Profiling Service Module. Before you use the parameter recommendations, verify that you have identified a node or grid and the requirement is to configure the node or grid optimally to run profiles.

The following configuration parameters have an impact on different components of the profiling and discovery installation. You can increase or decrease the values for these parameters based on the performance requirements:

Parameter	Description
Maximum Execution Pool Size	The maximum number of requests that the Data Integration Service can run concurrently. Requests include data previews, mappings, and profiling jobs. This parameter has an impact on the Data Integration Service.
Maximum Profile Execution Pool Size	The total number threads to run profiles. This parameter has an impact on the Profiling Service Module.
Temporary Directories	Location of temporary directories for the Data Integration Service process on the node. This parameter has an impact on the Data Integration Service machine.

Note: For more information about Data Integration Service parameters, see the *Profiling and Discovery Sizing Guidelines* article.

Data Integration Service Profiling Tuning Parameters

The following table lists the values for the Data Integration Service profiling tuning parameters:

Tuning Parameter	Value for 100,000 or lesser Rows with Row Sampling Enabled	Value for 100,000 or more Rows with Row Sampling Disabled	Description
Maximum Concurrent Columns	100	5	Maximum number of columns merged in single mapping.
Maximum Column Heap Size	512	512	Maximum cache size allocated for a mapping.
AdvancedProfilingServiceOptions.ColumnsThreshold	100	Default	Maximum number of columns merged for a mapping. Applicable only for column profiling.
AdvancedProfilingServiceOptions.DomainMappingColumnThreshold	100	50	Maximum number of columns merged for a mapping. Applicable only for data domain discovery.
AdvancedProfilingServiceOptions.MaxDataDomains	150	Default	Maximum number of data domains combined for a data domain discovery mapping.
ExecutionContextOptions.Optimizer.MaxMappingPorts	-1	Default	Controls platform optimization during mapping execution.

Tuning Parameter	Value for 100,000 or lesser Rows with Row Sampling Enabled	Value for 100,000 or more Rows with Row Sampling Disabled	Description
ExecutionContextOptions.Optimizer.PushdownType	none	Default	Controls pushdown optimization during mapping execution.
ExecutionContextOptions.Optimizer.ExtractSourceConstraints	false	Default	Controls extracting constraint details for a relational source during mapping execution.
ExecutionContextOptions.JVMOption1	1024M	1024M	Controls mapping process heap size.
ExecutionOptions.MaxProcessLifeTime	1800000	Default	Controls mapping process life time.
ExecutionContextOptions.Optimizer.NativeExpressionTx	false	Default	Controls SISR optimization during mapping execution.

Data Integration Service Node Scaling for Profiling

The execution time and throughput for data discovery varies based on the following parameters. It also depends on the hardware configuration of the Data Integration Service node.

- Volume of data in the tables and files
- Number of rows
- Number of unique or duplicate data sets
- Number of columns
- Column length
- Sampling options
- Number of data domains

You can choose a configuration from the following options to scale the Data Integration Service node for profiling:

- **Option 1 — Vertical hardware scaling with a standalone node**
  - This option is recommended if you have a single profiling resource.
  - The throughput for a single profiling resource can be increased by vertically scaling the node hardware configuration.
  - To achieve vertical scaling, you must calculate the Data Integration Service concurrency parameters for profiling and increase the number of CPU cores and system memory on the node.
- **Option 2 — Horizontal hardware scaling with multiple nodes**
  - This option is recommended if you have multiple profiling resources.
  - The throughput for multiple profiling resources can be increased by horizontally scaling the node hardware configuration. This configuration is used to increase the throughput across resources.
  - Create multiple nodes with dedicated profiling warehouse schema that can be selected at the resource level under the domain connection settings.

- **Option 3 — Grid configuration**
  - This option is used to increase the throughput for a single profiling resource by horizontal scaling.
  - Using this configuration, you can improve the data discovery throughput by up to 1.4x by increasing the total hardware capacity by 2x across multiple grid nodes.
  - The generation of mappings and compilation is performed on the master node. The execution takes place on both the master and the secondary nodes. Therefore, the hardware utilization on the master node is always higher than the secondary nodes.
  - The minimum hardware requirement recommended for a node used in a grid configuration is 32 CPU cores and 64 GB memory.

Data Integration Service Grid Configuration for Enterprise Data Catalog

In Enterprise Data Catalog, to use the Data Integration Service on a grid, you need to configure a shared directory with read and write access on all the nodes in the grid.

To configure the `<Informatica installation directory>/<profile_temp_folder>` shared directory on all the nodes, perform the following steps:

1. Log in to Informatica Administrator.
2. Navigate to **Manage** > **Services and Nodes** > **Domain Navigator**.
3. Select the grid in the **Domain Navigator**.
4. In the **Properties** > **Custom Properties** section, click the **Edit** icon.
5. In the **Edit Custom Properties** dialog box, click **New**.
6. In the **New Custom Property** dialog box, enter the following details:
   a. In the **Name** field, enter **AdvancedProfilingServiceOptions.CustomFlag8**.
   b. In the **Value** field, enter the temporary folder. For example, `/data2/profile_temp`.
7. Recycle the Data Integration Service.

In Enterprise Data Catalog, you can use the shared directory to perform the following tasks:

- To stage XML or JSON schemas on a Data Integration Service node
- To stage unstructured files and results on a Data Integration Service node
- To stage similarity results on a Data Integration Service node

Tuning Performance for Unique Key Inference Process

The unique key inference throughput and execution time are exponential and they depend on the following factors:

- Schema size
- Data set volume
- Profiling sample size
- Profiling concurrency

To tune unique key inference performance, configure the **ExecutionContextOptions.JVMoption1 = -Xmx4096M** JVM option in the **Informatica Administrator** > **Manage** > **Services and Nodes** > **Data Integration Service** > **Custom Properties** section.

Data Integration Service concurrency parameters

To increase the profiling performance throughput, you can tune the following Data Integration Service concurrency parameters:

- Maximum On-Demand Execution Pool Size
- Maximum Profile Execution Pool Size
- Maximum Concurrent Profile Jobs
- Maximum Concurrent Profile Threads
- AdvancedProfilingServiceOptions.FileConnectionLimit. This parameter applies to SAP, XML, and JSON sources.
- AdvancedProfilingServiceOptions.HiveConnectionLimit. This parameter applies to Hive sources.
- Maximum # of Connections for source connection

You can configure the concurrency parameters in one of the following ways:

- In the Administrator tool, navigate to the Data Integration Service properties to configure the concurrency parameters.
- In the command prompt, navigate to the `<Informatica installation directory>/isp/bin/plugins/ps/concurrencyParameters/` location.
  The *ConcurrencyParameters.properties* file in this location contains the default values of the Data Integration Service concurrency parameters for each load type. Run the *setConcurrencyParameters* script with the load type option to update the parameter values simultaneously for Data Integration Service.

Calculate Concurrency Parameter Values

You can calculate the value for concurrency parameters based on the number of cores. It is recommended that you multiply the number of cores with 0.6 and round off the value to the nearest whole value. For example, if the number of cores configured for the Data Integration Service is 16, then the concurrency parameter value is calculated as follows:16*0.6=9.6. The concurrency parameter value is rounded off to the nearest whole value, namely, 10.

The following table describes how to calculate the Data Integration Service concurrency parameters for profiling based on the number of cores:

Concurrency Parameter	Value*	Description
Maximum On-Demand Execution Pool Size	Number of cores x 0.6	Maximum parallel mappings run by the Data Integration Service that includes all types of mappings.
Maximum Profile Execution Pool Size	Number of cores x 0.6	Maximum parallel profiling mappings run by the Data Integration Service.
Maximum Concurrent Profile Jobs	Number of cores x 0.6	Maximum parallel tables or files processed.
Maximum Concurrent Profile Threads	Number of cores x 0.6	Maximum parallel mappings run by the Data Integration Service for File sources.
AdvancedProfilingServiceOptions.FileConnectionLimit (For CSV, XML, and JSON metadata sources)	Number of cores x 0.6	Maximum parallel profiling mappings run by the Data Integration Service for File sources.

Concurrency Parameter	Value*	Description
AdvancedProfilingServiceOptions.HiveConnectionLimit (For Hive metadata sources)	Number of cores x 0.6	Maximum parallel profiling mappings run by the Data Integration Service for Hive sources.
Maximum # of Connections	Number of cores x 0.6	Total number of database connections for each resource.
* Indicates the value of the parameter if the number of rows is less than or greater than 100, 000 rows with sampling of rows enabled or disabled.		

setConcurrencyParameters Script

You can run the script to update all the parameter values in the *ConcurrencyParameters.properties* file simultaneously. After you run the script, the script stops the Data Integration Service, updates the parameter values in the Data Integration Service, and then enables the Data Integration Service.

The *Maximum # Connection Pool Size* parameter is configured at the data source level. Therefore, you need to update this parameter in the Administrator tool for relational sources.

The setConcurrencyParameters script uses the following syntax:

```
setConcurrencyParameters.sh
[small/medium/large]
[DomainName]
[DISName]
[Username]
[Password]
```

The following table describes the setConcurrencyParameters script arguments:

Argument	Description
small/medium/large	Required. Enter small, medium, or large.
DomainName	Required. Enter the domain name.
DISName	Required. Enter the Data Integration Service name.
Username	Required. Enter the user name that the Data Integration Service uses to access the Model Repository Service.
Password	Required. Enter the password for the Model repository user.

Usage

```
./setConcurrencyParameters.sh [small/medium/large] [DomainName] [DISName] [Username]
[Password]
```

Example

```
./setConcurrencyParameters.sh small domain DIS Administrator Administrator
```

Default Parameter Values for Small, Medium, and Large Options in the Script

Parameter	Small	Medium	Large
Data Integration Service server hardware	CPU cores is16. Memory is 32 GB.	CPU cores is 32. Memory is 64 GB.	CPU cores is 64. Memory is 128 GB.
Maximum On-Demand Execution Pool Size	10	20	40
Maximum Profile Execution Pool Size	10	20	40
Maximum Concurrent Profile Jobs	10	20	40
Maximum Concurrent Profile Threads	10	20	40
AdvancedProfilingServiceOptions.FileConnectionLimit (For SAP, XML, JSON sources)	10	20	40
AdvancedProfilingServiceOptions.HiveConnectionLimit (For Hive sources)	10	20	40
Maximum # of Connections for source connection	10	20	40

Tuning profiling warehouse parameters

The profiling warehouse stores profiling results. More than one Profiling Service Module may point to the same profiling warehouse. The main resource for the profiling warehouse is disk space.

The following table lists the parameters that you need to tune to improve the performance of the profiling warehouse based on the size of data:

Parameter	Low	Medium	High	Description
Tablespace	50 GB	200 GB	500 GB	The maximum amount of disk space required to temporarily store the profiling results before these results get stored in the Enterprise Data Catalog.
CPU	4	8	16	The number of cores required by the profiling warehouse

Note: For more information about the profiling warehouse guidelines, see the Profiling and Discovery Sizing Guidelines article.

Tuning performance for profiling on Spark engine

To increase the profiling throughput for high volume sources, run the profiles on the Spark engine in the Hadoop environment.

To optimize profiling performance, run the following resources on the Spark engine:

- HDFS sources
- Hive sources
- Partitioned parquet files on Amazon S3 and Microsoft Azure Data Lake Store Gen2.

To tune the profiling throughput, configure the following parameters:

- Increase the value of **spark.yarn.executor.memoryOverhead** from 384 to 2048.
- Configure the **spark.dynamicAllocation.maxExecutors** parameter to 5 for every 3 nodes. For example, you can configure **spark.dynamicAllocation.maxExecutors** to 10 for 6 nodes.

To configure the parameters, navigate to the **Informatica Administrator** > **Manage** > **Connections** > **Hadoop CCO connection** > **Spark Configuration** > **Advanced Properties** section.

The minimum system requirement for each node in the Spark cluster is 32 CPU cores and 128 GB of memory. The following table describes the Service Level Agreement (SLA) to process metadata extraction and profiling:

Load Type	Number of Cluster Nodes	Hive Schema Size	Number of Tables per Schema	Number of Columns per Table	Recommended Sampling	Number of Data Domains	SLA
Medium	3	Upto 10	100	100	1 million	10	48 - 72 Hours
High	6	Upto 100	200	100	10 million	10	1 - 4 Weeks

Load Type	Number of Cluster Nodes	Hive Schema Size	Number of Tables per Schema	Number of Columns per Table	Recommended Sampling	Number of Data Domains	SLA
Very High	12	Upto 100	1000	100	10 million	10	4 - 10 Weeks

Note:
- The SLA is based on Hive table execution performance on an on-premise HDP/CDH cluster.
- The recommended mode for profiling is random percentage sampling. The maximum recommended sampling for the Spark engine is 10 million. You can extrapolate the sampling based on the actual row count.
- The profiling execution time and throughput will vary based on whether the data source is cloud based or resides on a local file system. It also depends on the following factors:
  - Data source type
  - Volume of content in the tables or files
  - Number of tables, rows, and columns
  - Column length
  - File type
  - Average and maximum file size
  - Number of unique and duplicate datasets
  - Sampling mode
  - Number of data domains
  - Source system hardware configuration
  - Disk latency
  - Network latency between source systems
- The data domain discovery execution time is linearly proportional to the total number of data domains selected for profiling.
- The cluster performance will vary based on the underlying hardware configuration of the cluster nodes and the cluster type such as EMR/HDP/CDP.

For information about best practices and to tune the Spark engine, see the Tuning and Sizing Guidelines for Informatica® Data Engineering Integration 10.4.0 and Performance Tuning for the Spark Engine articles.

Memory sizing to run profiles on large unstructured files

You can run a profile on unstructured files, such as PDF, Microsoft Excel, and Microsoft Word files.

Memory Requirements to Run Profiles on Large PDF Data Sources

Large unstructured files in PDF format require more memory during profiling.

The following table lists the memory that you can configure for the *ExecutionContextOptions.JVMOption1* parameter based on the PDF file size:

PDF File Size	Memory (in MB)
Less than or equal to 25 MB	1024M
Less than or equal to 50 MB	2048M
Less than or equal to 100 MB	5120M

Memory Requirements to Run Profiles on Large Microsoft Excel or Microsoft Word Files

The following table lists the memory that you can configure for the *ExecutionContextOptions.JVMOption1* parameter based on the Microsoft Excel or Microsoft Word file size:

File Size	Memory (in MB)
Less than or equal to 5 MB	1024M
Less than or equal to 25 MB	2048M
Less than or equal to 100 MB	7168M

To tune the *ExecutionContextOptions.JVMOption1* parameter, navigate to the **Manage** > **Services and Nodes** > **Data Integration Service** > **Custom Properties** section in Informatica Administrator.

Resource configuration parameters for profiling

Configure the profiling parameters using custom properties to optimize the profiling performance. You can also configure JVM options at the resource level in the Catalog Administrator tool.

Snowflake Resource

To optimize the profiling performance for Snowflake high volume resources, configure the following custom properties in Informatica Administrator:

- **AdvancedProfilingServiceOptions.ColumnsThreshold**. Set the value to 10.
- **ExecutionContextOptions.JVMoption1**. Set the value to -Xmx2048M.

To configure these parameters, navigate to the **Services and Nodes** > **Data Integration Service** > **Custom Properties** section.

Microsoft Azure Data Lake Store Gen2 Resource

You can configure the following parameters to optimize the profiling performance for Microsoft Azure Data Lake Store Gen2 resources:

Heap Size for pmdtmsvc2 Process

To set the maximum heap size for pmdtmsvc2 process used for profiling Microsoft Azure Data Lake Store Gen2 high volume resources in the native runtime environment, configure the following custom property :

`ExecutionContextOptions.JVMoption1` and set the value to -Xmx4096M.

The pmdtmsvc2 process memory requirement might increase for a large file. When the configured heap size does not meet the file processing memory requirement, a heap size error appears and the mapping is marked as failed.

To configure the custom property, navigate to the **Informatica Administrator** > **Services and Nodes** > **Data Integration Service** > **Custom Properties** section.

Concurrent Threads

You can set concurrent threads to download the files from Microsoft Azure Data Lake Store Gen2 source. By default, this option is set to 10. It is recommended to configure this value in the range of current number of cores to twice the number of CPU cores. For example, on a 16 CPU cores server, you can configure this parameter from 16 through 32.

You can configure the following option at resource level to override the default value:

```
-Dconnector.access.concurrent.thread
```

To configure this JVM option at the resource level in Catalog Administrator, open the resource and set the custom option on the **Metadata Load Settings** tab.

Block Size

You can set the block size to download the files in chunks using multiple threads. By default, the option is set to 8MB.

You can configure this option at resource level to override the default value:

```
-Dconnector.access.block.sizeInMB
```

It is recommended that you consider the pmdtmsvc2 process heap size and concurrent threads to configure this option to an optimum value to avoid redundant network calls.

To configure this JVM option at the resource level in Catalog Administrator, open the resource and set the custom option on the **Metadata Load Settings** tab.

Heap Size

Configure the Data Integration Service heap size to 10GB in Informatica Administrator before you run the data domain discovery profiles in Catalog Administrator. You can increase this value if the data domains exceed 100.

Cassandra Resource

You can configure the following parameters to optimize the profiling performance for Cassandra resources:

Heap Size for pmdtmsvc2 Process

Configure the ExecutionContextOptions.JVMoption1 custom property to -Xmx2048M to optimize the profiling performance for Cassandra high volume resources. This parameter sets the maximum heap size for pmdtmsvc2 process used for profiling.

To configure these parameters, navigate to **Informatica Administrator** > **Services and Nodes** > **Data Integration Service** > **Properties** > **Custom Properties** section.

Heap Size

Configure the Data Integration Service heap size to 10 GB when you run data domain discovery on Cassandra resources. Increase the heap size if the number of data domains exceed 100.

Profiling large tables on JDBC connections

Profiling large tables in MySQL, PostgreSQL, Vertica, Athena, and Databricks Delta using JDBC connectivity might cause performance issues as sampling is not supported on the native execution engine.

To resolve the issue, you can limit the number of rows fetched by the queries. Configure the following option at the resource level to specify the custom data profiling sampling value to 1000 and limit the number of rows to 1000: `-DcustomSampling='limit 1000'`.

To configure this option at the resource level in Catalog Administrator, open the resource and set the option on the **Metadata Load Settings** tab in the **Custom Options** property.

Profiling recommendations for a Databricks cluster

Verify the following recommendations to optimally run profiling on a Databricks cluster:

- The Databricks cluster and source data ,must reside in the same geographical region to reduce network latency.
- For better throughput and performance, you can enable autoscaling.

Hardware Recommendations for Databricks Runtime 5.5 LTS

Verify the following hardware recommendations:

- You can configure the worker or driver type node as Standard_DS3_v3 that includes 128.0 GB of memory, 32 CPU cores, and 6 DBU.
- You can configure a minimum of 1 worker node and a maximum of 8 worker nodes with autoscaling enabled.

Profiling Recommendation for Cloud Data Warehouse Resources

The following load type tables list the maximum number of assets you can profile from cloud data warehouse resources based on the configured resource memory:

Low

The scanner memory configured is 2 GB.

Resource Type	Data Type	Data Size	Enterprise Data Catalog Asset Count
Google BigQuery	Table	2K	100K
Amazon S3 schema merge (JSON/XML/CSV)	File	10 GB	NA

Medium

The scanner memory configured is 4 GB.

Resource Type	Data Type	Data Size	Enterprise Data Catalog Asset Count
Google BigQuery	Table	4K	200K
Amazon S3 schema merge (JSON/XML/CSV)	File	20 GB	NA

High

The scanner memory configured is 12 GB.

Resource Type	Data Type	Data Size	Enterprise Data Catalog Asset Count
Google BigQuery	Table	8K	400K
Amazon S3 schema merge (JSON/XML/CSV)	File	50 GB	NA

CHAPTER 5

Content Migration

This chapter includes the following topic:

- Catalog export and migration recommendations, 52

Catalog export and migration recommendations

As part of upgrade to version 10.5.1, you must export, back up, and migrate the catalog data. You must also back up the Model Repository and upgrade the Model Repository Service with the backed up content as part of the upgrade process. Verify the recommendations when you upgrade to version 10.5.1.
Detailed instructions about content backup, export, and migration are included in *Upgrading from Versions 10.4.0 and Later*

The following sections list the recommendations for upgrade:

Model Repository Backup and Upgrade Recommendations

See the following table for the heap size recommendation based on the size of the Model Repository backup file:

Model Repository Backup File Size	Recommended Heap Size
200 MB	4 GB
200 MB-2 GB	8 GB
2 GB-4 GB	12 GB

You can configure the Model Repository Service heap size from Informatica Administrator. Access the Model Repository Service **Properties** > **Advanced Properties** section from Informatica Administrator and set the recommended heap size based on the file size.

Catalog Data Backup and Export Recommendations

See the following table for the heap size recommendation to back up and export catalog data:

Asset Count	Recommended Heap Size	Approximate Time for Data Backup in Hours
2 million	10 GB	0.5
10 million	10 GB	1
20 million	10 GB	2
50 million	15 GB	5
100 million	20 GB	10

The expected throughput for backup and export is 10 million assets per hour.

For a medium load type, the default memory configuration is sufficient for the export of 20 million assets with the Informatica domain node configured with 32 GB of memory. If the Informatica domain node is shared by multiple applications, such as Data Privacy Management, Enterprise Data Catalog, and Enterprise Data Preparation, you can restrict the maximum heap memory used by the migration process. You can use the `java-Xmx<memory>m-DskipReadOnlyCheck=true -jar export.jar export` option with the export utility as shown in the following sample to restrict the maximum heap memory: `java-Xmx10240m-DskipReadOnlyCheck=true -jar export.jar export`.

For a high load type, the default memory configuration is sufficient for the export of 50 million assets with the Informatica domain node configured with 64 GB of memory. If the Informatica domain node is shared by multiple applications, such as Data Privacy Management, Enterprise Data Catalog, and Enterprise Data Preparation, you can restrict the maximum heap memory used by the migration process. You can use the `java-Xmx<memory>m-DskipReadOnlyCheck=true -jar export.jar export` option with the export utility as shown in the following sample to restrict the maximum heap memory: `java-Xmx15360m-DskipReadOnlyCheck=true -jar export.jar export`.

Catalog Data Migration Recommendations

See the following table for the heap size recommendation and the estimated time to migrate data based on the number of assets in the catalog:

Asset Count	Recommended Heap Size	Approximate Time for Data Migration in Hours
2 million	6 GB	1.3
10 million	10 GB	3
20 million	10 GB	6
50 million	15 GB	15
100 million	20 GB	30

The expected throughput for migration is 2-4 million assets per hour.

For a medium load type, the default memory configuration is sufficient for the migration of 20 million assets with the Informatica domain node configured with 32 GB of memory. If the Informatica domain node is shared by multiple applications, such as Data Privacy Management, Enterprise Data Catalog, and Enterprise

Data Preparation, you can restrict the maximum heap memory used by the migration process. You can set the `xdocs.content.systemProperties=` variable to `-Xmx10G` in the `<Informatica installation directory>/services/CatalogService/Binaries/MigrationModuleConfigurations.properties` file to restrict the maximum heap memory.

For a high load type, the default memory configuration is sufficient for the migration of 50 million assets with the Informatica domain node configured with 64 GB of memory. If the Informatica domain node is shared by multiple applications, such as Data Privacy Management, Enterprise Data Catalog, and Enterprise Data Preparation, you can restrict the maximum heap memory used by the migration process. You can set the `xdocs.content.systemProperties=` variable to `-Xmx15G` in the `<Informatica installation directory>/services/CatalogService/Binaries/MigrationModuleConfigurations.properties` file to restrict the maximum heap memory.

Catalog re-index estimates

After you migrate the catalog data, and you observe missing data links, you can re-index the catalog to resolve the issue.
See the following table for an approximate re-indexing time based on the volume of data in the catalog:

Asset Count in Catalog Data	Approximate Re-indexing Time in Hours
2 million	0.30
10 million	1
20 million	2
50 million	5
100 million	10

If the catalog data includes a high volume of shared links, it is recommended to increase the publish task memory from the default value of 4 GB to 8 GB. You can use the `LdmCustomOptions.ingest.publish.task.memory.mb` Catalog Service custom property to increase the default publish task memory value.

CHAPTER 6

Performance Tuning Parameters

This chapter includes the following topic:

- Performance tuning parameters, 55

Performance tuning parameters

You can tune the Enterprise Data Catalog performance by configuring the custom properties for the Catalog Service. Also, you can configure custom options for each resource. The following section lists the custom properties and parameters that you can use:

Configure the scanner memory

You can increase the memory allocated to resources for specific scenarios where the default value needs to be increased. Use the following parameters to increase the memory for each load type.

Data Size	Custom property for the Catalog Service	Custom option for the resource	Default value
Low	`LdmCustomOptions.scanner.memory.low`	`-Dscanner.memory.low`	2048 MB
Medium	`LdmCustomOptions.scanner.memory.medium`	`-Dscanner.memory.medium`	4096 MB
High	`LdmCustomOptions.scanner.memory.high`	`-Dscanner.memory.high`	12288 MB

Improve ingestion performance for high volume sources

You can improve ingestion performance for high volume sources using custom parameters. Verify that adequate amount of free memory is available on all the cluster nodes.

Configure the following custom property for the Catalog Service:
`LdmCustomOptions.ingest.resolve.task.memory.mb`

Configure the following custom option for the resource: `-Dingest.resolve.task.memory.mb`

Default is 4096 MB.

Improve publishing performance for high volume sources

You can improve publishing performance for high volume sources using custom parameters. Verify that adequate amount of free memory is available on all cluster nodes.

Configure the following custom property for the Catalog Service:
`LdmCustomOptions.ingest.publish.task.memory.mb`

Configure the following custom option for the resource: `-Dingest.publish.task.memory.mb`

Default is 4096 MB.

Improve purge performance for high volume sources

You can improve purge performance for high volume sources using custom parameters. Verify that adequate amount of free memory is available on all cluster nodes.

Configure the following custom property for the Catalog Service:
`LdmCustomOptions.ingest.purge.task.memory.mb`

Configure the following custom option for the resource: `-Dingest.purge.task.memory.mb`

Default is 4096 MB.

Improve publish batch size performance for high volume sources

You can improve publish batch size performance for high volume sources using custom parameters.

Attention: Use the parameter with caution as increasing the batch size might result in CPU throttling that might impact overall performance and product stability.

Configure the following custom property for the Catalog Service:
`LdmCustomOptions.ingest.publish.object.batch.size`

Configure the following custom option for the resource: `-Dingest.publish.object.batch.size`

Default is 1000.

Tune concurrent jobs per node

You can increase the number of concurrent jobs running on a node using a custom property.

Attention: The CPU and memory requirements per node increases if you increase the value for this parameter, which might lead to stability and performance issues with the MongoDB database. The memory that you configure for the node must be based on the MongoDB WiredTiger cache memory associated with the MongoDB license.

Configure the following custom property for the Catalog Service:
`LdmCustomOptions.orchestration.exec.engine.concurrent.jobs`

Default is 1 per node.

Tune concurrent file processors

You can increase the number of concurrent file processors.

Attention: Increasing the value might lead to CPU throttling issues and thread blocks.

Configure the following custom option for the resource: `-DfileProcessorCount`

Default is 2.

Tune concurrent threads to download a file from a data lake

You can increase the number of concurrent threads to download files from a data lake in a high volume scenario.

To prevent redundant network calls, it is recommended to tune this value based on the values specified for the pmdtmsvc2 process heap size and concurrent threads.

Configure the following custom option for the resource: `-Dconnector.access.block.sizeInMB`

Default is 8 MB.

Tune the maximum number of Apache Solr documents stored before auto commit

You govern the maximum number of Apache Solr documents stored in-memory before an auto commit is performed.

Configure the following custom option for the resource: `-Dsolrconfig.updatehandler.autocommit.maxdocs`

Default is 1000.

Enable resource debugging

You can use custom parameters to generate log files for resource debugging.

Attention: Use the parameter as required as it might impact disk utilization for high volume resources.

You must add the parameter to the custom properties in the EDCConfiguration.properties file under the following directory: `<Informatica installation directory>/services/CatalogService/Binaries/`

Set the number of log files for the resources using the orchestration.job.log.max.file = <value> parameter.

Set the log file size for the resources using the orchestration.job.log.max.file.size = <value> parameter.

To validate the details, you must verify the Nomad configuration for each job from the log configuration: Job > TaskGroup > Task > logConfig

Configure the following custom option for the resource: `-Dscannerloglevel-`

Skip connection properties test

You can skip testing the connection properties configured for a resource using a custom option. To skip testing the connection properties, configure the following custom option for the resource: `-DskipTestConnection`

Skip metadata extraction

You skip extracting metadata for a resource using a custom option. To skip extracting metadata, configure the following custom option for the resource: `-DskipExtraction`

Configure resources in similarity discovery group

You can use a custom property to increase the number of resources in the similarity resource group.

Attention: Similarity computation is a resource-intensive process and might require additional disk space based on the volume of data.

For all high volume load types, it is recommended to increase the number of resources per resource group only if additional hardware is available.

Note: You must restart the Catalog Service after you configure this parameter.

Configure the following custom property for the Catalog Service: `LdmCustomOptions.scanner.simDiscoveryScanner.maxPermittedResourceCount`

Default is 4.

Tune similarity computation and throughput

You can increase the similarity throughput by adding PostgreSQL database hardware. You can then tune the throughput using the property. The default value of 5 represents 8 CPU cores and 16 GB of memory for a PostgreSQL node. To increase the count by 5, you must add an addition 4 CPU cores and 4 GB of memory for the PostgreSQL node.

Note: You must restart the Catalog Service if you configure this parameter.

Configure the following custom property for the Catalog Service:
`LdmCustomOptions.scanner.simDiscoveryScanner.partitionPairJobCount`

Default is 5.

Configure MongoDB WiredTiger cache memory

You can control the percentage of memory allocated for MongoDB WiredTiger cache.

Note: This parameter is only applicable for 10.5.0.1 and later versions. Based on the license, you can allocate a maximum of 24 GB of memory per node for the MongoDB WiredTiger cache. To increase the memory allocation, contact Informatica Global Customer Support. To configure the MongoDB WiredTiger cache memory, go to **ICS MongoDB Service Options > Mongo Custom Options > MemoryLimitPercentage**

Default is 40.

Configure the number of seeds for MongoDB query

To configure the lineage, you can split the MongoDB query in batches of seeds instead of a single query for all seeds.

Attention: Use the parameter with caution as it is set at the global level and might impact overall lineage performance and stability.

Configure the following custom property for the Catalog Service:
`LdmCustomOptions.ldm.lineage.max.seed.per.mongo.query`

Default is 1.

Configure the number of iterations for MongoDB query

To configure the lineage, you can split the MongoDB query in batches of seeds for every iteration.

Configure the following custom property for the Catalog Service:
`LdmCustomOptions.ldm.lineage.default.depth.for.iteration`

Default is 30.

Specify the maximum file size for unstructured files

You can specify the maximum file size for large unstructured files that can be scanned.

Configure the following custom property for the Catalog Service:
`LdmCustomOptions.scanner.fileScanner.unstructuredMaxFileSizeInMB`

Attention: Use the parameter with caution as it is set at the global level and might impact overall lineage performance and stability.

Default is 15 MB.

Configure access filters

You can use this parameter to configure the resource batch size for each Apache Solr request after you update the access filters for the resource.

Configure the following custom property for the Catalog Service:
`LdmCustomOptions.ldm.max.resource.for.search.query`

Default is 700.

Improve the connection assignment performance

The connection assignment tasks run with the memory option configured for the resource. To improve the connection assignment performance for high volume sources, you can override the memory allocated to the resource using custom parameters.

Configure the following custom property for the Catalog Service:
`LdmCustomOptions.ingest.assign.task.memory.mb-`

Configure the following custom option for the resource: `-Dingest.assign.task.memory.mb`

If you encounter `OutOfMemory` errors for PowerCenter resources enabled for reference resources, you can resolve the issue by setting the `LdmCustomOptions.ingest.use.new.lineage.publish.imp` custom property to `false`.

Improve the Composite Data Domain Discovery performance

You can set the following options for the custom property `LdmCustomOptions.orchestration.nomad.exec.additional.jvm.flags` in the Catalog Service to improve the performance of Composite Data Domain Discovery:

LDM Custom Options	Description
-Dscanner.CDDInferenceResultFetcher.threads	Set the property to edit the number of result fetcher threads based on the inference from the Data Domains. Minimum value is 1. Default is 4.
-Dscanner.CDDScanner.customPageSize	Set the property to edit the page size for API calls that import columns with data domains. Maximum value is 10000. Default is 100.
-Dscanner.CDDScanner.skipUnstructuredFiles	Set the property to true to skip the Composite Data Discovery processing for unstructured files. Default is false.
-Dscanner.CDDScanner.threads	Set the property to edit the number of parallel threads for Composite Data Discovery. Minimum value is 1. Default is 4.

Performance tuning parameters for Data Asset Analytics

Performance tuning involves configuring parameters for the Data Asset Analytics repository database.

Tuning Oracle Database Parameters for Data Asset Analytics

The machine that hosts the Oracle database must meet the following minimum recommended system requirements:

Environment Size Number of Assets	Requirement
Small Up to 5 million	- 8 CPU cores - 24 GB memory
Medium 5 - 20 million	- 8 CPU cores - 24 GB memory
Large 20 – 50 million	- 8 CPU cores - 24 GB memory
Extra large 50 – 100 million	- 12 CPU cores - 24 GB memory

The following table lists the database parameters that you can configure to tune the Data Asset Analytics performance for a small size environment with up to 5 million assets:

Parameter	Value
sga_target	8 GB
pga_aggregate_target	4 GB
pga_aggregate_limit	8 GB

The following table lists the database parameters that you can configure to tune the Data Asset Analytics performance for a medium size environment with 5 - 20 million assets:

Parameter	Value
sga_target	8 GB
pga_aggregate_target	4 GB
pga_aggregate_limit	8 GB

The following table lists the database parameters that you can configure to tune the Data Asset Analytics performance for a large size environment with 20 – 50 million assets:

Parameter	Value
sga_target	8 GB
pga_aggregate_target	4 GB
pga_aggregate_limit	8 GB

The following table lists the database parameters that you can configure to tune the Data Asset Analytics performance for an extra large size environment with 50 – 100 million assets:

Parameter	Value
sga_target	10 GB
pga_aggregate_target	8 GB
pga_aggregate_limit	16 GB

Configure additional database parameters as shown in the following table:

Parameter	Value
filesystemio_options	NONE or ASYNC
processes	1000

Parameter	Value
open_cursors	1000
optimizer_capture_sql_plan_baselines	False
optimizer_use_sql_plan_baselines	False

Database Configuration Guidelines

Consider the following guidelines when you configure the machine that hosts the database:

- The network latency between the machine on which Data Asset Analytics is installed and the database must be in the range of 20-30 milliseconds.
- You must configure a minimum value of 50 GB for the temporary tablespace for the database.
- You must configure a minimum of three online redo log files with a size of 4 GB each.
- It is recommended that you enable periodic collection of table statistics to ensure optimum performance.
- When the number of assets and concurrent users increase, you must scale up the parameter values and hardware configuration based on the resource usage.

Tuning Microsoft SQL Server Database Parameters for Data Asset Analytics

The machine that hosts the SQL Server database must meet the following minimum recommended system requirements:

Environment Size Number of Assets	Requirement
Small Up to 5 million	- 8 CPU cores - 20 GB memory
Medium 5 - 20 million	- 8 CPU cores - 20 GB memory
Large 20 – 50 million	- 12 CPU cores - 20 GB memory
Extra large 50 – 100 million	- 24 CPU cores - 20 GB memory

Perform the following steps to tune Data Asset Analytics performance:

1. Set the value of the cost threshold for parallelism parameter to 5.
2. Run the following command to set the value of the maximum degree of parallelism parameter to 5: `ALTER DATABASE SCOPED CONFIGURATION SET MAXDOP = 5.`
3. Run the following command to set the value of the `BATCH_MODE_ON_ROWSTORE` parameter to OFF: `ALTER DATABASE SCOPED CONFIGURATION SET BATCH_MODE_ON_ROWSTORE = OFF.`

   Note: This parameter is only supported on SQL Server 2019 and later versions.

Database Configuration Guidelines

Consider the following guidelines when you configure the machine that hosts the database:

- The network latency between the machine on which Data Asset Analytics is installed and the database must be less than 60 milliseconds.
- The disk response time for the database machine must be less than 50 milliseconds.
- It is recommended that you enable periodic collection of table statistics to ensure optimum performance.
- When the number of assets and concurrent users increase, you must scale up the parameter values and hardware configuration based on the resource usage.

Tuning Microsoft Azure SQL Database Parameters for Data Asset Analytics

The following table lists the recommended minimum system requirements for the machine that hosts the SQL Server database on Microsoft Azure:

Environment Size Number of Assets	Microsoft Azure Service Tier	Minimum Requirement
Small Up to 5 million	vCore - General Purpose (Gen5)	vCores – 8, Storage – 5 GB
	DTU - Standard	S7: Number of DTUs – 800, Storage – 20 GB
Medium 5 – 20 Million	vCore - General Purpose (Gen5)	vCores – 32, Storage – 30 GB
	DTU - Standard	S9: Number of DTUs – 1600, Storage – 30 GB
Large 20 – 50 Million	vCore - General Purpose (Gen5)	vCores – 32, Storage – 100 GB
	DTU - Standard	S9: Number of DTUs – 1600, Storage – 70 GB
Extra large 50 – 100 Million	vCore - General Purpose (Gen5)	vCores – 40, Storage – 100 GB
	DTU - Standard	Not Recommended
	DTU - Premium	P15: Number of DTUs – 4000, Storage – 100 GB

Run the following command to configure the value of the maximum degree of parallelism parameter to 5:
`ALTER DATABASE SCOPED CONFIGURATION SET MAXDOP = 5.`

Database Configuration Guidelines

Consider the following guidelines when you configure the machine that hosts the database:

- The network latency between the machine on which Data Asset Analytics is installed and the database must be less than 60 milliseconds.
- It is recommended that you enable periodic collection of table statistics to ensure optimum performance.
- When the number of assets and concurrent users increase, you must scale up the parameter values and hardware configuration based on the resource usage.

Tuning Amazon Web Services RDS for SQL Server Database Parameters

The following table lists the recommended minimum system requirements for the machine that hosts the SQL Server database on Amazon Web Services RDS:

Environment Size Number of Assets	Requirements
Small Up to 5 million	- 8 vCPU - General Purpose SSD storage - 20 GB disk size
Medium 5 – 20 million	- 16 vCPU - General Purpose SSD storage - 30 GB disk size
Large 20 – 50 million	- 16 vCPU - General Purpose SSD storage - 100 GB disk size
Extra large 50 – 100 million	- 16 vCPU - Provisioned IOPS storage - 2000 IOPS - 150 GB disk size

Run the following command to configure the maximum degree of parallelism to 5: `ALTER DATABASE SCOPED CONFIGURATION SET MAXDOP = 5.`

Database Configuration Guidelines

Consider the following guidelines when you configure the machine that hosts the database:

- The network latency between the machine on which Data Asset Analytics is installed and the database must be less than 60 milliseconds.
- It is recommended that you enable periodic collection of table statistics to ensure optimum performance.
- When the number of assets and concurrent users increase, you must scale up the parameter values and hardware configuration based on the resource usage.

Tuning PostgreSQL Database Parameters for Data Asset Analytics

If you configured PostgreSQL as the repository database for Data Asset Analytics, perform the following steps to tune Data Asset Analytics performance:

- If you have 50 million or lesser number of assets in the catalog, configure the PostgreSQL parameters as shown in the following table:

Parameter	Value
work_mem	4 GB
shared_buffers	16 GB
effective_cache_size	15 GB
effective_io_concurrency	200

Parameter	Value
min_wal_size	2 GB
max_wal_size	8 GB
max_worker_processes	16
max_parallel_workers_per_gather	4
max_parallel_workers	16
max_parallel_maintenance_workers	4
random_page_cost	1.1
max_locks_per_transaction	1024

Note: If the catalog includes more than 50 million assets, you can tune the parameters based on your requirement.

- Add the `pg_hint_plan` extension in the PostgreSQL database to use existing `hints` in SQL statements and generate the optimal execution plan.
- Add the `BatchMechanism=COPY` parameter in the JDBC connection string as shown in the following sample: `jdbc:Informatica:postgresql://<host name>:<port number>;DatabaseName=<DB Name>;BatchMechanism=COPY`

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